

Primitive Transition Chain Law

Autonomous discussion Pass 110 studies the primitive vectors obtained by clearing barycentric transition denominators.

For finite supports $S \subset T$, define

$$L_{S,T} = \text{lcm}(|S|, |T|), \quad \eta_{S,T} = L_{S,T} \tau_{S,T}.$$

If $|S| = n$, $|T| = m$, and $g = \text{gcd}(n, m)$, then $\eta_{S,T}$ has entries $(m - n)/g$ on S and $-n/g$ on $T \setminus S$. It is an integral primitive zero-sum vector in K_T .

For a chain $S \subset T \subset U$, let

$$C = \text{lcm}(L_{S,T}, L_{T,U}, L_{S,U}).$$

Multiplying the rational coboundary identity by C gives the exact integral rescaling law

$$\frac{C}{L_{S,T}} e_{T,U} \eta_{S,T} + \frac{C}{L_{T,U}} \eta_{T,U} = \frac{C}{L_{S,U}} \eta_{S,U}.$$

Thus primitive transition vectors do not usually compose strictly. Strict primitive composition occurs only in equal-conductor cases; otherwise the coefficients $C/L_{A,B}$ are essential. The useful support-edge datum is therefore the weighted pair $(L_{S,T}, \eta_{S,T})$, equivalently the rational transition $\tau_{S,T}$, not the primitive line alone.

This mirrors the primitive repair torsor issue. The rescaled chain law is closed in the additive kernel K_U , but the common-conductor sum can be a nonprimitive multiple of the endpoint vector. Primitivity is an arithmetic normalization inside additive kernel data, not a sub-cocycle condition.

The checker `artifacts/reports/pass110-primitive-transition-chain-law-check.json` verifies edge-vector primitivity, the weighted common-conductor chain law, the strict-composition classification, failure of primitive lines to be sufficient in general, and compatibility with the Pass-107 repair-torsor warning.

The next task is to incorporate the Claude Code MacNeille reflection checker review: add the non-lattice witness, correct or separate the antitone completion closure rule, and record reflected/principal-unreflected classifications.